

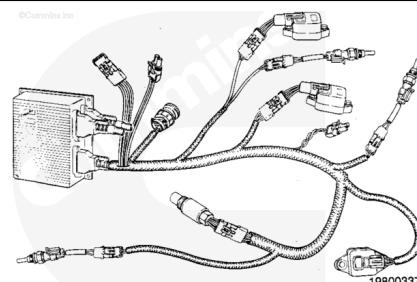
Trainings ▾

General Information

Heavy-Duty Only

To communicate with the Centinel™ control module, the public datalink circuit is used for an electronic service tool.

Note : No service tool is provided for Heavy-Duty applications. The Centinel™ control module will use the public datalink.



For Celect™ Plus engines, the datalink is powered and uses a 6-pin Deutsch connector. (For Celect™ engines, the public datalink is accessed by the control harness through splices in the cab.) The wiring positions follow:

Position A - Datalink (+)

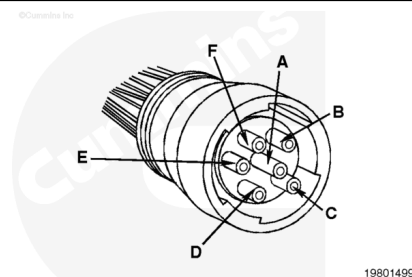
Position B - Datalink (-)

Position C - Battery (12 or 24 VDC)

Position D - Open

Position E - Block ground

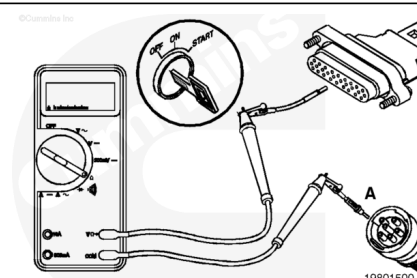
Position F - Not used.



Resistance Check

⚠ CAUTION ⚠

Use test lead, Part Number 3822758, on the Centinel™ control module connector and use test lead, Part Number 3823993, on the 6-pin Deutsch connector to avoid damage to the connector pins.



Disconnect the harness connector from the Centinel™ control module and the Deutsch 6-pin connector.

Turn the keyswitch to the "OFF" position.

Measure the resistance from pin 8 (Heavy-Duty) or pin 6 (High-Horsepower) of the Centinel™ control module harness connector to pin A of the 6-pin Deutsch connector (or the proper wire splice for the Celect installation).

The multimeter **must** show a closed circuit (10 ohms or less).

If the circuit is **not** closed, repair or replace the Centinel™ harness. Refer to Procedure 019-131

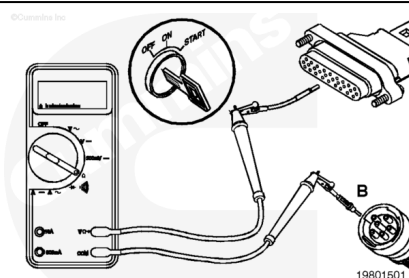
(/qs3/pubsys2/xml/en/procedures/96/96-019-131-tr.html).

Measure the resistance from pin 9 (Heavy-Duty) or pin 8 (High-Horsepower) of the Centinel™ control module harness connector to pin B of the 6-pin Deutsch connector (or the proper wire splice for the Celect™ installation).

The multimeter **must** show a closed circuit (10 ohms or less).

If the circuit is **not** closed, repair or replace the Centinel™ harness. Refer to Procedure 019-131

(/qs3/pubsys2/xml/en/procedures/96/96-019-131-tr.html).



Check for Short Circuit to Ground

⚠ CAUTION ⚠

Use test lead, Part Number 3823993, for the 6-pin Deutsch connector to avoid damage to the connector pins.

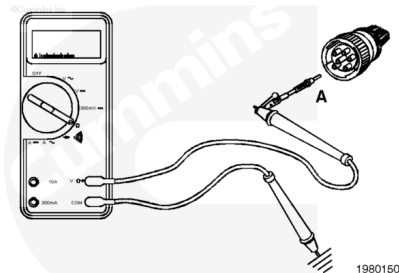
Disconnect the harness connector from the Centinel™ control module.

Measure the resistance from pin A of the Deutsch connector or pin 8 of the Centinel™ control module harness connector to the engine block.

The multimeter **must** show an open circuit (1M ohms or more).

If the circuit is **not** open, repair or replace the Centinel™ harness. Refer to Procedure 019-131

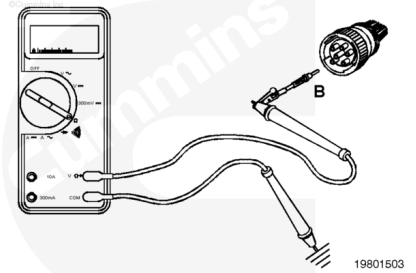
(/qs3/pubsys2/xml/en/procedures/96/96-019-131-tr.html).



Measure the resistance from pin B of the 6-pin Deutsch connector or pin 9 of the Centinel™ control module harness connector to the engine block. The multimeter **must** show an

open circuit (1M ohms or more).

If the circuit is **not** open, repair or replace the Centinel™ harness. Refer to Procedure 019-131 (</qs3/pubsys2/xml/en/procedures/96/96-019-131-tr.html>).

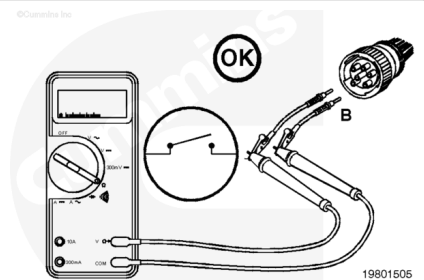


Check for Short Circuit from Pin to Pin

Deutsch

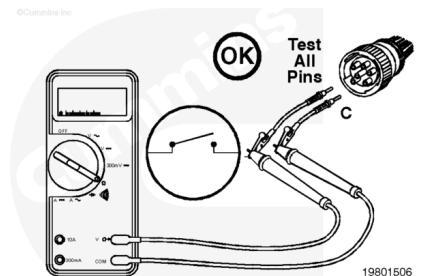
Measure the resistance from pin B to all other pins in the connector. The multimeter **must** show an open circuit (1M ohms or more).

If the circuit is **not** open, repair or replace the Centinel™ harness. Refer to the appropriate base engine troubleshooting and repair manual.



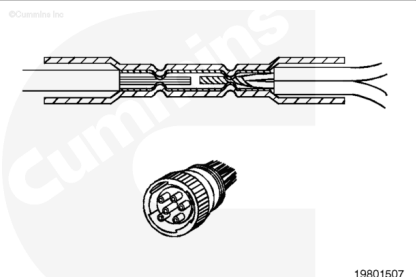
Measure the resistance from pin C to all other pins in the connector. The multimeter **must** show an open circuit (1M ohms or more).

If the circuit is **not** open, repair or replace the Centinel™ harness. Refer to the appropriate base engine troubleshooting and repair manual.



Voltage Check

Locate the datalink connector on the Centinel™ harness. The location will depend on the installation procedures.

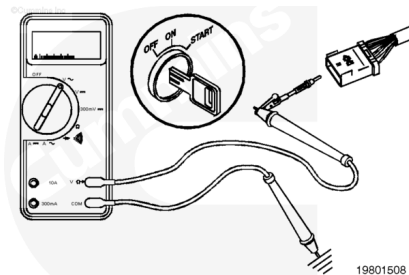


Use test lead, Part Number 3823993, for the 6-pin Deutsch connector to avoid damage to the connector pins.

Turn the keyswitch ON.

Turn the dial on the multimeter to measure DC voltage.

Measure the voltage from pin 8 of the Centinel™ control module harness connector to the engine block. The multimeter **must** show 2.5 to 5 VDC.



Measure the voltage from pin 9 of the Centinel™ control module harness connector to the engine block. The multimeter **must** show 0 to 2.5 VDC.

